

Note Set # 3

The complete free space solution to $\nabla^2 \mathbf{E} + k^2 \mathbf{E} = 0$ is

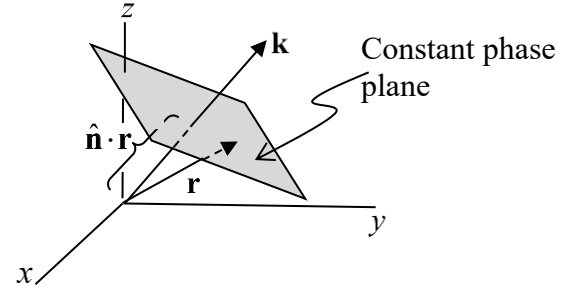
$$\mathbf{E}(\mathbf{r}) = \mathbf{E}_0 e^{-j\mathbf{k} \cdot \mathbf{r}} = \mathbf{E}_0 e^{-jk\hat{\mathbf{n}} \cdot \mathbf{r}}$$

where

$$\mathbf{r} = x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}$$

is a position vector of the observation point (x, y, z)

$$\begin{aligned}\mathbf{k} &= k_x \hat{\mathbf{x}} + k_y \hat{\mathbf{y}} + k_z \hat{\mathbf{z}} \\ &= k \hat{\mathbf{n}}\end{aligned}$$



$\mathbf{k} \doteq$ wave vector

So that expressions (and calculations) can be simplified we will define the wave vectors $\mathbf{k}_i$, $\mathbf{k}_r$ and $\mathbf{k}_t$ for a plane wave incident on a boundary as follows:

$$\begin{aligned}\mathbf{k}_i &= k_1 \hat{\mathbf{n}}_i = -k_1 \sin \theta_i \hat{\mathbf{y}} + k_1 \cos \theta_i \hat{\mathbf{z}} = -k_{1y} \hat{\mathbf{y}} + k_{1z} \hat{\mathbf{z}} \\ \mathbf{k}_r &= k_1 \hat{\mathbf{n}}_r = -k_1 \sin \theta_i \hat{\mathbf{y}} - k_1 \cos \theta_i \hat{\mathbf{z}} = -k_{1y} \hat{\mathbf{y}} - k_{1z} \hat{\mathbf{z}} \\ \mathbf{k}_t &= k_2 \hat{\mathbf{n}}_t = -k_2 \sin \theta_t \hat{\mathbf{y}} + k_2 \cos \theta_t \hat{\mathbf{z}} = -k_{2y} \hat{\mathbf{y}} + k_{2z} \hat{\mathbf{z}}\end{aligned}$$

where

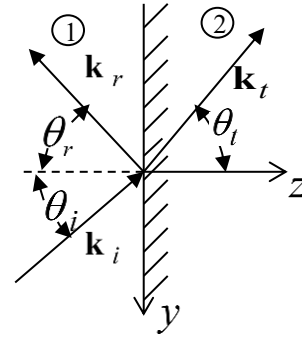
$$\begin{aligned}k_{1y} &\equiv k_1 \sin \theta_i \\ k_{1z} &\equiv k_1 \cos \theta_i \\ k_{2y} &\equiv k_2 \sin \theta_t \\ k_{2z} &\equiv k_2 \cos \theta_t\end{aligned}$$

For example

$$\begin{aligned}e^{-j\mathbf{k}_i \cdot \mathbf{r}} &= e^{-j(-k_{1y}\hat{\mathbf{y}} + k_{1z}\hat{\mathbf{z}}) \cdot \mathbf{r}} \\ &= e^{jk_{1y}y - jk_{1z}z} = e^{jk_1 y \sin \theta_i - jk_1 z \cos \theta_i}\end{aligned}$$

$$e^{-j\mathbf{k}_r \cdot \mathbf{r}} = e^{jk_{1y}y + jk_{1z}z} = e^{jk_1 y \sin \theta_i + jk_1 z \cos \theta_i}$$

$$e^{-j\mathbf{k}_t \cdot \mathbf{r}} = e^{jk_{2y}y - jk_{2z}z} = e^{jk_2 y \sin \theta_t - jk_2 z \cos \theta_t}$$



Or wave unit vectors in terms of the direction cosines:

$$\hat{\mathbf{n}}_i = \cos \beta^i \hat{\mathbf{y}} + \cos \gamma^i \hat{\mathbf{z}} = -\sin \theta_i \hat{\mathbf{y}} + \cos \theta_i \hat{\mathbf{z}}$$

$$\hat{\mathbf{n}}_r = \cos \beta^i \hat{\mathbf{y}} + \cos \gamma^i \hat{\mathbf{z}} = -\sin \theta_i \hat{\mathbf{y}} - \cos \theta_i \hat{\mathbf{z}},$$

$$\hat{\mathbf{n}}_t = \cos \beta^t \hat{\mathbf{y}} + \cos \gamma^t \hat{\mathbf{z}} = -\sin \theta_t \hat{\mathbf{y}} + \cos \theta_t \hat{\mathbf{z}}$$

○ Plane wave reflection and diffraction at a boundary.

known:

$$\mathbf{E}_i^a, \text{ frequency } \omega, \hat{\mathbf{n}}, \theta_i$$

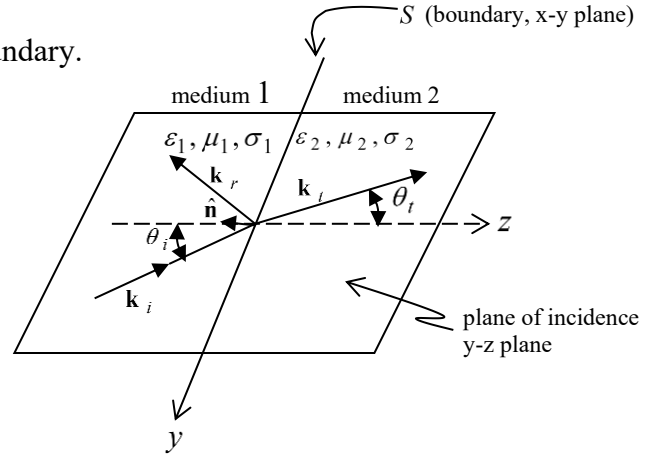
$$\mu_1, \epsilon_1, \sigma_1, \mu_2, \epsilon_2, \sigma_2$$

$$\theta_r = \theta_i$$

unknown:

$$\mathbf{E}_r^a, \mathbf{E}_t^a, \theta_t$$

construct:



$$\mathbf{E}_i = \mathbf{E}_i^a e^{-j\mathbf{k}_i \cdot \mathbf{r}} = \mathbf{E}_i^a e^{jk_1 y \sin \theta_i - jk_1 z \cos \theta_i}$$

$$\mathbf{E}_i = \mathbf{E}_i^a e^{-j\mathbf{k}_i \cdot \mathbf{r}} = \mathbf{E}_i^a e^{jk_1 y \sin \theta_i - jk_1 z \cos \theta_i}$$

$$\mathbf{E}_r = \mathbf{E}_r^a e^{-j\mathbf{k}_r \cdot \mathbf{r}} = \mathbf{E}_r^a e^{jk_1 y \sin \theta_i + jk_1 z \cos \theta_i}$$

$$\mathbf{E}_r = \mathbf{E}_i^a \frac{E_r^a}{E_i^a} e^{-j\mathbf{k}_r \cdot \mathbf{r}} = \mathbf{E}_r^a \Gamma e^{jk_1 y \sin \theta_i + jk_1 z \cos \theta_i}$$

$$\mathbf{E}_t = \mathbf{E}_t^a e^{-j\mathbf{k}_t \cdot \mathbf{r}} = \mathbf{E}_t^a e^{jk_2 y \sin \theta_t - jk_2 z \cos \theta_t}$$

$$\mathbf{E}_t = \mathbf{E}_i^a \frac{E_t^a}{E_i^a} e^{-j\mathbf{k}_t \cdot \mathbf{r}} = \mathbf{E}_i^a T e^{jk_2 y \sin \theta_t - jk_2 z \cos \theta_t}$$

$$\mathbf{H}_i = \frac{j}{\omega \mu_1} \nabla \times \mathbf{E}_i = \frac{1}{\omega \mu_1} \mathbf{k}_i \times \mathbf{E}_i$$

$$\mathbf{H}_r = \frac{j}{\omega \mu_1} \nabla \times \mathbf{E}_r = \frac{1}{\omega \mu_1} \mathbf{k}_r \times \mathbf{E}_r$$

$$\mathbf{H}_t = \frac{j}{\omega \mu_2} \nabla \times \mathbf{E}_t = \frac{1}{\omega \mu_2} \mathbf{k}_t \times \mathbf{E}_t$$

From the boundary conditions on the tangential $\mathbf{E}$ and $\mathbf{H}$ components.

$$\hat{\mathbf{n}} \times (\mathbf{E}_1 - \mathbf{E}_2) = 0$$

$$\hat{\mathbf{n}} \times (\mathbf{E}_i + \mathbf{E}_r) = \hat{\mathbf{n}} \times \mathbf{E}_t$$

so at $z = 0$ (boundary)

$$\hat{\mathbf{n}} \times \mathbf{E}_1|_{z=0} = \hat{\mathbf{n}} \times \mathbf{E}_2|_{z=0}$$

$$\hat{\mathbf{n}} \times (\mathbf{E}_i^a e^{jk_1 y \sin \theta_i} + \mathbf{E}_r^a e^{jk_1 y \sin \theta_i}) = \hat{\mathbf{n}} \times \mathbf{E}_t^a e^{jk_2 y \sin \theta_t}$$

$$\underbrace{\hat{\mathbf{n}} \times (\mathbf{E}_i^a + \mathbf{E}_r^a)}_{\text{amplitude}} \underbrace{e^{jk_1 y \sin \theta_i}}_{\text{phase}} = \underbrace{\hat{\mathbf{n}} \times \mathbf{E}_t^a}_{\text{amplitude}} \underbrace{e^{jk_2 y \sin \theta_t}}_{\text{phase}}$$

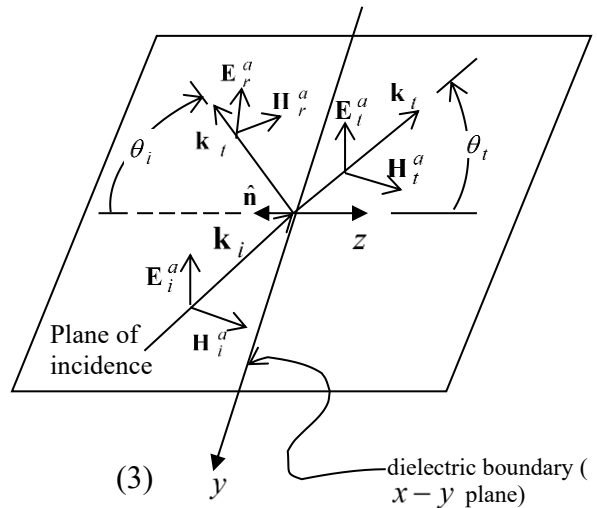
Equate phases:

$$e^{jk_1 y \sin \theta_i} = e^{jk_2 y \sin \theta_t} \Rightarrow k_1 \sin \theta_i = k_2 \sin \theta_t$$

Snell's law

$$\begin{aligned} (e^{\mathbf{j}\mathbf{k}_i \cdot \mathbf{r}} = e^{\mathbf{j}\mathbf{k}_t \cdot \mathbf{r}})|_{z=0} &\Rightarrow (\mathbf{k}_i \cdot \hat{\mathbf{y}} = \mathbf{k}_t \cdot \hat{\mathbf{y}}) \\ &\Rightarrow \mathbf{k}_{i,\text{tang}} = \mathbf{k}_{t,\text{tang}} \end{aligned}$$
$$\hat{\mathbf{n}} \times (\mathbf{E}_i^a + \mathbf{E}_r^a) = \hat{\mathbf{n}} \times \mathbf{E}_t^a \quad (1)$$
$$\begin{aligned}\hat{\mathbf{n}} \times (\mathbf{H}_1 - \mathbf{H}_2) &= 0 \\ \hat{\mathbf{n}} \times (\mathbf{H}_i - \mathbf{H}_r) &= \hat{\mathbf{n}} \times \mathbf{H}_t\end{aligned}$$
$$\hat{\mathbf{n}} \times \left(\frac{1}{\omega \mu_1} \mathbf{k}_i \times \mathbf{E}_i^a + \frac{1}{\omega \mu_1} \mathbf{k}_r \times \mathbf{E}_r^a \right) = \hat{\mathbf{n}} \times \left(\frac{1}{\omega \mu_2} \mathbf{k}_t \times \mathbf{E}_t^a \right)$$
$$\hat{\mathbf{n}} \times \mathbf{k}_i \times \mathbf{E}_i^a + \hat{\mathbf{n}} \times \mathbf{k}_t \times \mathbf{E}_r^a = \frac{\mu_1}{\mu_2} \hat{\mathbf{n}} \times \mathbf{k}_t \times \mathbf{E}_t^a \quad (2)$$

- Case I $\mathbf{E}_i$ normal to the plane of incidence ($\perp$ polarization): $\mathbf{E}_i^a = E_{i\perp} \hat{x}$

$$\mathbf{k}^{(i)} \cdot \mathbf{E}^a = 0 \quad \text{and} \quad \hat{\mathbf{n}} \cdot \mathbf{E}^a = 0$$


- Take $\hat{\mathbf{n}} \times (1)$ and apply the identity
- $$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \cdot \mathbf{C})\mathbf{B} - (\mathbf{A} \cdot \mathbf{B})\mathbf{C}$$

$$\hat{\mathbf{n}} \times (1) = \hat{\mathbf{n}} \times \hat{\mathbf{n}} \times (\mathbf{E}_i^a + \mathbf{E}_r^a) = \hat{\mathbf{n}} \times \hat{\mathbf{n}} \times \mathbf{E}_t^a$$

$$\cancel{(\hat{\mathbf{n}} \cdot \overset{0}{\nearrow} \mathbf{E}_i^a)} \hat{\mathbf{n}} - \cancel{(\hat{\mathbf{n}} \cdot \hat{\mathbf{n}})} \mathbf{E}_i^a + \cancel{(\hat{\mathbf{n}} \cdot \overset{0}{\nearrow} \mathbf{E}_r^a)} \hat{\mathbf{n}} - \cancel{(\hat{\mathbf{n}} \cdot \hat{\mathbf{n}})} \overset{1}{\nearrow} \mathbf{E}_r^a$$

$$= (\hat{\mathbf{n}} \cdot \mathbf{E}_t^a) \hat{\mathbf{n}} - (\hat{\mathbf{n}} \cdot \hat{\mathbf{n}}) \mathbf{E}_t^a$$

$$\mathbf{E}_i^a + \mathbf{E}_r^a = \mathbf{E}_t^a$$

which are all vectors in the θ_i -dir. ($\perp$ plane of incidence)

$$\Rightarrow E_{i\perp} + E_{r\perp} = E_{t\perp} \quad (4)$$

- Apply (3) to (2)

$$\begin{aligned} (\hat{\mathbf{n}} \cdot \mathbf{E}_i^a) \hat{\mathbf{k}}_i - (\hat{\mathbf{n}} \cdot \hat{\mathbf{k}}_i) \mathbf{E}_i^a + (\hat{\mathbf{n}} \cdot \mathbf{E}_r^a) \hat{\mathbf{k}}_r - (\hat{\mathbf{n}} \cdot \hat{\mathbf{k}}_r) \mathbf{E}_r^a \\ = \frac{\mu_1}{\mu_2} \left[(\hat{\mathbf{n}} \cdot \mathbf{E}_t^a) \hat{\mathbf{k}}_t - (\hat{\mathbf{n}} \cdot \hat{\mathbf{k}}_t) \mathbf{E}_t^a \right] \end{aligned}$$

$$(\hat{\mathbf{n}} \cdot \hat{\mathbf{k}}_i) \mathbf{E}_i^a + (\hat{\mathbf{n}} \cdot \hat{\mathbf{k}}_r) \mathbf{E}_r^a = \frac{\mu_1}{\mu_2} (\hat{\mathbf{n}} \cdot \hat{\mathbf{k}}_t) \mathbf{E}_t^a$$

Since the coefficients (dot products) are all scalars and since the $\mathbf{E}$'s are all in the same (x) direction, we can drop the x unit vector on the field.

$$(\hat{\mathbf{n}} \cdot \mathbf{k}_i) E_{i\perp} + (\hat{\mathbf{n}} \cdot \mathbf{k}_r) E_{r\perp} = \frac{\mu_1}{\mu_2} (\hat{\mathbf{n}} \cdot \mathbf{k}_i) E_{t\perp} \quad (5)$$

- Combine (4) and (5)

$$\frac{E_{r\perp}}{E_{i\perp}} = \frac{-\mu_2 \hat{\mathbf{n}} \cdot \mathbf{k}_i + \mu_1 \hat{\mathbf{n}} \cdot \mathbf{k}_t}{\mu_2 \hat{\mathbf{n}} \cdot \mathbf{k}_r - \mu_1 \hat{\mathbf{n}} \cdot \mathbf{k}_t} \doteq \Gamma_{\perp} \quad (6a)$$

$$\frac{E_{t\perp}}{E_{i\perp}} = \frac{\mu_2 \hat{\mathbf{n}} \cdot \mathbf{k}_i - \mu_2 \hat{\mathbf{n}} \cdot \mathbf{k}_r}{-\mu_2 \hat{\mathbf{n}} \cdot \mathbf{k}_r + \mu_1 \hat{\mathbf{n}} \cdot \mathbf{k}_t} \doteq T_{\perp} \quad (6b)$$

Substitute $\hat{\mathbf{n}} = -\hat{\mathbf{z}}$ and the values of $\mathbf{k}_i$, $\mathbf{k}_r$, and $\mathbf{k}_t$ from the beginning of this set of notes into (6a) and (6b) to get

$$\Gamma_{\perp} = \frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t} \quad (7a)$$

$$T_{\perp} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t} \quad (7b)$$

Duality

$$\mathbf{E} \rightarrow \mathbf{H}$$

$$\mathbf{H} \rightarrow -\mathbf{E}$$

$$\left. \begin{array}{l} \mu \rightarrow \varepsilon \\ \varepsilon \rightarrow \mu \end{array} \right\} \Rightarrow \begin{array}{l} k \rightarrow k \\ \eta \rightarrow 1/\eta \end{array}$$

$$\mathbf{M} \rightarrow -\mathbf{J}$$

$$\mathbf{J} \rightarrow \mathbf{M}$$

Example: $\nabla \times \mathbf{E} = -j\omega\mu\mathbf{H} - \mathbf{M} \rightarrow \nabla \times (\mathbf{H}) = -j\omega(\varepsilon)(-\mathbf{E}) - (-\mathbf{J})$
 $\nabla \times \mathbf{H} = j\omega\varepsilon\mathbf{E} + \mathbf{J}$

Case 2 Use duality to find the $\parallel$ reflection and transmission coefficients.

$$\begin{aligned} \Gamma_{\perp} &= \frac{E_{r\perp}}{E_{i\perp}} = \frac{-\mu_2 \hat{\mathbf{n}} \cdot \mathbf{k}_i + \mu_1 \hat{\mathbf{n}} \cdot \mathbf{k}_t}{\mu_2 \hat{\mathbf{n}} \cdot \mathbf{k}_r - \mu_1 \hat{\mathbf{n}} \cdot \mathbf{k}_t} \rightarrow \frac{H_{r\parallel}}{H_{i\parallel}} = \frac{-\varepsilon_2 \hat{\mathbf{n}} \cdot \mathbf{k}_i + \varepsilon_1 \hat{\mathbf{n}} \cdot \mathbf{k}_t}{\varepsilon_2 \hat{\mathbf{n}} \cdot \mathbf{k}_r - \varepsilon_1 \hat{\mathbf{n}} \cdot \mathbf{k}_t} \\ &\downarrow \\ &\frac{-E_{r\parallel} / \eta_1}{E_{i\parallel} / \eta_1} = \frac{-\varepsilon_2 \hat{\mathbf{n}} \cdot \mathbf{k}_i + \varepsilon_1 \hat{\mathbf{n}} \cdot \mathbf{k}_t}{\varepsilon_2 \hat{\mathbf{n}} \cdot \mathbf{k}_r - \varepsilon_1 \hat{\mathbf{n}} \cdot \mathbf{k}_t} \\ &\frac{E_{r\parallel}}{E_{i\parallel}} = \frac{\varepsilon_2 \hat{\mathbf{n}} \cdot \mathbf{k}_i - \varepsilon_1 \hat{\mathbf{n}} \cdot \mathbf{k}_t}{\varepsilon_2 \hat{\mathbf{n}} \cdot \mathbf{k}_r - \varepsilon_1 \hat{\mathbf{n}} \cdot \mathbf{k}_t} \doteq \Gamma_{\parallel} \\ T_{\perp} &= \frac{E_{t\perp}}{E_{i\perp}} = \frac{\mu_2 \hat{\mathbf{n}} \cdot \mathbf{k}_i - \mu_2 \hat{\mathbf{n}} \cdot \mathbf{k}_r}{\mu_2 \hat{\mathbf{n}} \cdot \mathbf{k}_r + \mu_1 \hat{\mathbf{n}} \cdot \mathbf{k}_t} \rightarrow \frac{H_{t\parallel}}{H_{i\parallel}} = \frac{\varepsilon_2 \hat{\mathbf{n}} \cdot \mathbf{k}_i - \varepsilon_2 \hat{\mathbf{n}} \cdot \mathbf{k}_r}{-\varepsilon_2 \hat{\mathbf{n}} \cdot \mathbf{k}_r + \varepsilon_1 \hat{\mathbf{n}} \cdot \mathbf{k}_t} \\ &\downarrow \\ &\frac{Et_{i\parallel} / \eta_2}{E_{i\parallel} / \eta_1} = \frac{\varepsilon_2 \hat{\mathbf{n}} \cdot \mathbf{k}_i - \varepsilon_2 \hat{\mathbf{n}} \cdot \mathbf{k}_r}{-\varepsilon_2 \hat{\mathbf{n}} \cdot \mathbf{k}_r + \varepsilon_1 \hat{\mathbf{n}} \cdot \mathbf{k}_t} \\ &\frac{E_{t\parallel}}{E_{i\parallel}} = \frac{\eta_2}{\eta_1} \left(\frac{\varepsilon_2 \hat{\mathbf{n}} \cdot \mathbf{k}_i - \varepsilon_2 \hat{\mathbf{n}} \cdot \mathbf{k}_r}{-\varepsilon_2 \hat{\mathbf{n}} \cdot \mathbf{k}_r + \varepsilon_1 \hat{\mathbf{n}} \cdot \mathbf{k}_t} \right) = T_{\parallel} \end{aligned}$$

Which, using $\hat{\mathbf{n}} = -\hat{\mathbf{z}}$ and the values of $\mathbf{k}_i$, $\mathbf{k}_r$, and $\mathbf{k}_t$ from the beginning of this set of notes reduce to:

$$\Gamma_{\parallel} = \frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i} \quad (8a)$$

$$T_{\parallel} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i} \quad (8b)$$

Important consequences of reflection from an engineering standpoint.

- Physical Optics approximation:

Normal incidence on a perfect conductor with $|\mathbf{E}_i| = E_x^a$

$$\Rightarrow \theta_i = 0^\circ, \mathbf{E}_2 = 0, \mathbf{H}_2 = 0, \theta_t = 0^\circ$$

The total magnetic field in air is therefore (pg. 3.6):

$$\mathbf{H}_1 = \mathbf{H}_i + \mathbf{H}_r = \frac{\hat{\mathbf{n}}_i}{\eta_1} \times E_x^a e^{-jk_1 \hat{\mathbf{n}}_i \cdot \mathbf{r}} \hat{\mathbf{x}} + \frac{\hat{\mathbf{n}}_r}{\eta_1} \times E_x^a \Gamma_{\perp} e^{-jk_1 \hat{\mathbf{n}}_r \cdot \mathbf{r}} \hat{\mathbf{x}}$$

$$\hat{\mathbf{n}}_i = -\sin \theta_i \hat{\mathbf{y}} + \cos \theta_i \hat{\mathbf{z}} = -\sin 0^\circ \hat{\mathbf{y}} + \cos 0^\circ \hat{\mathbf{z}} = \hat{\mathbf{z}}$$

$$\hat{\mathbf{n}}_i \cdot \mathbf{r} = \hat{\mathbf{z}} \cdot (x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}) = z$$

$$\hat{\mathbf{n}}_r = -\sin \theta_i \hat{\mathbf{y}} - \cos \theta_i \hat{\mathbf{z}} = -\sin 0^\circ \hat{\mathbf{y}} - \cos 0^\circ \hat{\mathbf{z}} = -\hat{\mathbf{z}}$$

$$\hat{\mathbf{n}}_r \cdot \mathbf{r} = -\hat{\mathbf{z}} \cdot (x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}) = -z$$

$$\begin{aligned} \Gamma &= \frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t} = \frac{\eta_2 \cos 0^\circ - \eta_1 \cos 0^\circ}{\eta_2 \cos 0^\circ + \eta_1 \cos 0^\circ} \\ &= \frac{\eta_2 / \eta_1 - 1}{\eta_2 / \eta_1 + 1}, \eta_2 = \sqrt{\frac{\mu_2}{\epsilon - j\sigma/\omega}} \rightarrow 0 \text{ as } \sigma \rightarrow \infty \\ &= -1 \end{aligned}$$

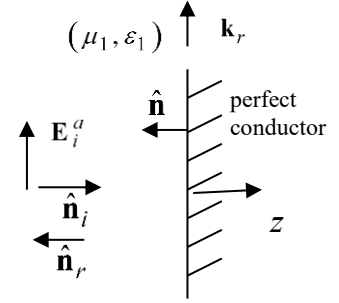
$$\begin{aligned} \mathbf{H}_1 &= \frac{\hat{\mathbf{z}}}{\eta_1} \times E_x^a e^{-jk_1 z} \hat{\mathbf{x}} + \frac{(-\hat{\mathbf{z}})}{\eta_1} \times E_x^a (-1) e^{-jk_1 (-z)} \hat{\mathbf{x}} \\ &= \left(\frac{E_x^a}{\eta_1} e^{-jk_1 z} + \frac{E_x^a}{\eta_1} e^{jk_1 z} \right) \hat{\mathbf{y}} = H_y^a e^{-jk_1 z} \hat{\mathbf{y}} + H_y^a e^{jk_1 z} \hat{\mathbf{y}} = 2H_y^a \cos(k_1 z) \hat{\mathbf{y}} \end{aligned}$$

$\therefore$ On the surface of the conductor which is at ($z = 0$)

$$\hat{\mathbf{n}} \times \left(\mathbf{H}_1 - \mathbf{H}_2 \right) \Big|_{z=0} = \mathbf{J}_s \rightarrow \hat{\mathbf{n}} \times (2H_y^a \hat{\mathbf{y}}) = \mathbf{J}_s \quad \text{or}$$

$2\hat{\mathbf{n}} \times \mathbf{H}^i = \mathbf{J}_s$

(the Physical Optics approximation)



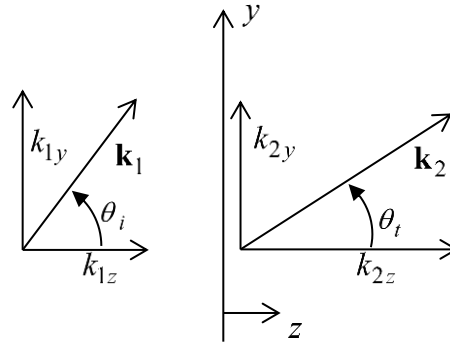
More important consequences of reflection

- Phase matching condition

$$k_{1y} = k_{2y}$$

$$k_1 \sin \theta_i = k_2 \sin \theta_t$$

(also called Snell's law)



The phases must match at the boundary because the velocities must be the same in the direction parallel to the boundary

$$v_{1y} = v_{2y}$$

$$\omega/k_{1y} = \omega/k_{2y} \Rightarrow k_{1y} = k_{2y}$$

- Brewster angle $\Gamma_{\parallel} = 0$ angle of total transmission

Example $\Gamma_{\parallel} = \frac{\varepsilon_2 k_{1z} - \varepsilon_1 k_{2z}}{\varepsilon_2 k_{1z} + \varepsilon_1 k_{2z}} = 0$ at $\theta_i = \theta_B$

$$\Rightarrow \varepsilon_2 k_{1z} - \varepsilon_1 k_{2z} = 0$$

or

$$\varepsilon_2 k_1 \cos \theta_B - \varepsilon_1 k_2 \cos \theta_t = 0 \quad (i)$$

Recall Snell's Law

$$k_1 \sin \theta_B = k_2 \sin \theta_t \quad (ii)$$

If $\mu_1 = \mu_2$ (i) and (ii) give

$$\theta_B = \tan^{-1} \sqrt{\varepsilon_2 / \varepsilon_1} \quad (iii)$$

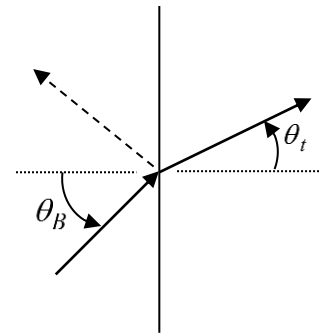
and (iii) can be written $\tan \theta_B = k_2 / k_1$

or $k_1 \sin \theta_B = k_2 \cos \theta_B \quad (iv)$

compare (ii) and (iv) $\Rightarrow \theta_B = \pm(\theta_t - 90^\circ)$

+ leads to $\theta_t = 90^\circ + \theta_B$ which is not possible

- leads to $\theta_t = 90^\circ - \theta_B$ which is physically possible (v)



Still more important consequences of reflection

- Critical angle – angle of total reflection

$$E_t \sim e^{-j\mathbf{k}_t \cdot \mathbf{r}}$$

$$\mathbf{k}_t \cdot \mathbf{r} = (-k_{2y}\hat{\mathbf{y}} + k_{2z}\hat{\mathbf{z}}) \cdot (x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}})$$

$$= -yk_2 \sin \theta_t + zk_2 \cos \theta_t$$

$$= -yk_2 \sin \theta_t + z\sqrt{k_2^2 - k_2^2 \sin^2 \theta_t} \quad (\text{from Snell's law})$$

$$= -yk_1 \sin \theta_i + z\sqrt{k_2^2 - k_1^2 \sin^2 \theta_i} \quad (\text{from Snell's law})$$

$$E_t \sim \underbrace{e^{jy k_1 \sin \theta_i}}_{\text{propagation in negative } \mathbf{k}_r \text{ direction}} \underbrace{e^{-jk_2 z \sqrt{1 - \frac{k_1^2}{k_2^2} \sin^2 \theta_i}}}_{\text{attenuation in the z-direction}}$$

propagation in
negative $\mathbf{k}_r$
direction

$$1 - \frac{k_1^2}{k_2^2} \sin^2 \theta_i > 0$$

$\Rightarrow$ Propagation in
positive z direction

$$1 - \frac{k_1^2}{k_2^2} \sin^2 \theta_i < 0$$

$\Rightarrow$ Attenuation in the z -direction

$$1 - \frac{k_1^2}{k_2^2} \sin^2 \theta_c = 0 \quad (\Rightarrow k_{2z} = 0)$$

$\Rightarrow$ No field in region 2

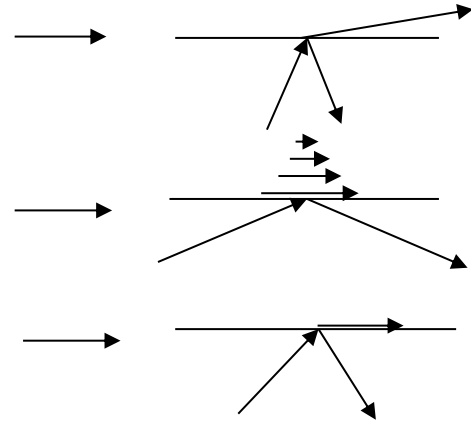
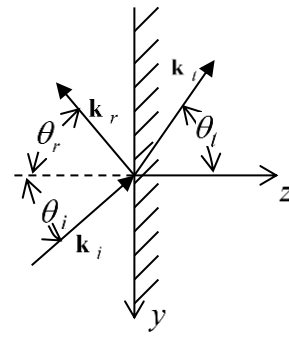
$$\text{so } \theta_c = \sin^{-1} \left(\frac{\sqrt{\mu_2 \epsilon_2}}{\sqrt{\mu_1 \epsilon_1}} \right) \text{ critical angle or angle of total reflection}$$

since usually $\mu_1 = \mu_2 = \mu_0$

$$\theta_c = \sin^{-1} \sqrt{\frac{\epsilon_2}{\epsilon_1}}$$

$$\Rightarrow \epsilon_1 > \epsilon_2$$

$$\text{Note: } \frac{k_1}{k_2} \sin \theta_c = 1 \Rightarrow \theta_t = 90^\circ$$



Power Conservation at a dielectric boundary

Define time power density $\mathcal{S}_{av} = \text{Re}\{\mathbf{S}\} = \frac{1}{2} \text{Re}\{\mathbf{E} \times \mathbf{H}^*\}$

At a planar boundary power must be conserved

$$\Rightarrow \hat{\mathbf{n}} \cdot (\mathcal{S}_{av,1} - \mathcal{S}_{av,2}) = 0 \quad , \hat{\mathbf{n}} \text{ into region 1}$$

or

$$\hat{\mathbf{n}} \cdot (\mathcal{S}_{av}^i + \mathcal{S}_{av}^r - \mathcal{S}_{av}^t) = 0$$

Example/ When a $\parallel$ polarized wave is incident at the critical angle $\theta_i \doteq \theta_c \Rightarrow \theta_t = 90^\circ$

$$\Gamma_{\parallel} = \frac{\varepsilon_2 k_{1z} - \varepsilon_1 k_{2z}}{\varepsilon_2 k_{1z} + \varepsilon_1 k_{2z}} = \frac{\varepsilon_2 k_1 \cos \theta_c - \varepsilon_1 k_2 \overset{0}{\cancel{\cos 90^\circ}}}{\varepsilon_2 k_1 \cos \theta_c + \varepsilon_1 k_2 \overset{0}{\cancel{\cos 90^\circ}}} = 1 \quad \text{or} \quad \begin{matrix} k_{2z} = 0 \\ k_{1y} = k_1 \sin \theta_c \\ k_{1z} = k_1 \cos \theta_c \end{matrix}$$

From page 3.1

$$k_1 \mathbf{n}_i = -k_1 \sin \theta_c \hat{\mathbf{y}} + k_1 \cos \theta_c \hat{\mathbf{z}}$$

$$\doteq -k_{1y} \hat{\mathbf{y}} + k_{1z} \hat{\mathbf{z}}$$

$$\Rightarrow \mathbf{n}_i = -\frac{k_{1y}}{k_1} \hat{\mathbf{y}} + \frac{k_{1z}}{k_1} \hat{\mathbf{z}}$$

$$\text{similarly } \mathbf{n}_r = -\frac{k_{1y}}{k_1} \hat{\mathbf{y}} - \frac{k_{1z}}{k_1} \hat{\mathbf{z}}$$

and from page 3.6

$$\mathbf{E}_i = -\hat{\mathbf{n}}_i \times E_{i\parallel} e^{-jk_1 \hat{\mathbf{n}}_i \cdot \mathbf{r}} \hat{\mathbf{x}}$$

$$\mathbf{E}_r = -\hat{\mathbf{n}}_r \times \Gamma_{\parallel} E_{i\parallel} e^{-jk_1 \hat{\mathbf{n}}_r \cdot \mathbf{r}} \hat{\mathbf{x}}$$

Combining these two equations gives

$$\therefore E_{1y} = -E_0 \frac{k_{1z}}{k_1} e^{-jk_1 \hat{\mathbf{n}}_i \cdot \mathbf{r}} + E_0 \frac{k_{1z}}{k_1} e^{-jk_1 \hat{\mathbf{n}}_r \cdot \mathbf{r}}$$

$$E_{1z} = -E_0 \frac{k_{1y}}{k_1} e^{-jk_1 \hat{\mathbf{n}}_i \cdot \mathbf{r}} - E_0 \frac{k_{1y}}{k_1} e^{-jk_1 \hat{\mathbf{n}}_r \cdot \mathbf{r}}$$

Also from page 3.6

$$H_{1x} = E_0 \frac{1}{\eta_1} e^{-jk_1 \hat{\mathbf{n}}_i \cdot \mathbf{r}} + E_0 \frac{1}{\eta_1} e^{-jk_1 \hat{\mathbf{n}}_r \cdot \mathbf{r}}$$

In region 2

$$T_{\parallel} = \frac{2\varepsilon_2(\eta_2/\eta_1)k_{1z}}{\varepsilon_2k_{1z} + \varepsilon_1k_{2z}} = 2\eta_2/\eta_1$$

$$E_{2y} = -E_0 \cos 90^\circ T_{\parallel} e^{-jk_2\hat{\mathbf{n}}_t \cdot \mathbf{r}} = 0$$

$$E_{2z} = -E_0 \sin 90^\circ T_{\parallel} e^{-jk_2\hat{\mathbf{n}}_t \cdot \mathbf{r}} = -E_0 \frac{2\eta_2}{\eta_1} e^{-jk_2\hat{\mathbf{n}}_t \cdot \mathbf{r}}$$

$$H_{2x} = E_0 \frac{1}{\eta_2} \left(\frac{2\eta_2}{\eta_1} \right) e^{-jk_2\hat{\mathbf{n}}_t \cdot \mathbf{r}} = \frac{2E_0}{\eta_1} e^{-jk_2\hat{\mathbf{n}}_t \cdot \mathbf{r}}$$

$$\mathcal{S}_{av} = \frac{1}{2} \text{Re} \left\{ \mathbf{E} \times \mathbf{H}^* \right\} \text{ Time ave. power flow}$$

$$\begin{aligned} \text{Re} \left\{ S_{1z}^i \right\} &= \frac{1}{2} \text{Re} \left\{ E_{1y}^i \hat{\mathbf{y}} \times H_{1x}^{i*} \hat{\mathbf{x}} \right\} \cdot \hat{\mathbf{z}} \\ &= + \frac{E_0^2}{2\eta_1} \frac{k_{1z}}{k_1} = \frac{E_0^2}{2\eta_1} \cos \theta_c \end{aligned}$$

$$\begin{aligned} \text{Re} \left\{ S_{1y}^i \right\} &= \frac{1}{2} \text{Re} \left\{ E_{1z}^i \hat{\mathbf{z}} \times H_{1x}^{i*} \hat{\mathbf{x}} \right\} \cdot \hat{\mathbf{y}} \\ &= - \frac{E_0^2}{2\eta_1} \frac{k_{1y}}{k_1} = - \frac{E_0}{2\eta_1} \frac{k_1 \sin \theta_c}{k_1} = - \frac{E_0^2}{2\eta_1} \sin \theta_c \end{aligned}$$

$$\text{Re} \left\{ S_{1z}^r \right\} = - \frac{E_0^2}{2\eta_1} \cos \theta_c$$

$$\text{Re} \left\{ S_{1y}^r \right\} = - \frac{E_0^2}{2\eta_1} \frac{k_{1y}}{k_1} = - \frac{E_0^2}{2\eta_1} \cos \theta_c$$

$$\text{Re} \left\{ S_{2z} \right\} = 0$$

$$\text{Re} \left\{ S_{2y} \right\} = -2E_0^2 \frac{\eta_2}{\eta_1^2} \quad (\text{same if } \theta_1 > \theta_c ?)$$

check power flow at boundary

$$\hat{\mathbf{n}} \cdot \mathcal{S}_{av}^i + \hat{\mathbf{n}} \cdot \mathcal{S}_{av}^r = \hat{\mathbf{n}} \cdot \mathcal{S}_{av}^t \text{ where } \hat{\mathbf{n}} = -\hat{\mathbf{z}}$$

$$-\hat{\mathbf{z}} \cdot \left[\overbrace{\text{Re} \left\{ S_{1z}^i \right\} \hat{\mathbf{z}} + \text{Re} \left\{ S_{1y}^i \right\} \hat{\mathbf{y}}}^{\mathcal{S}_{av}^i} + \overbrace{\text{Re} \left\{ S_{1z}^r \right\} \hat{\mathbf{z}} + \text{Re} \left\{ S_{1y}^r \right\} \hat{\mathbf{y}}}^{\mathcal{S}_{av}^r} + \overbrace{\text{Re} \left\{ S_{2z} \right\} \hat{\mathbf{z}} + \text{Re} \left\{ S_{2y} \right\} \hat{\mathbf{y}}}^{\mathcal{S}_{av}^t} \right]$$

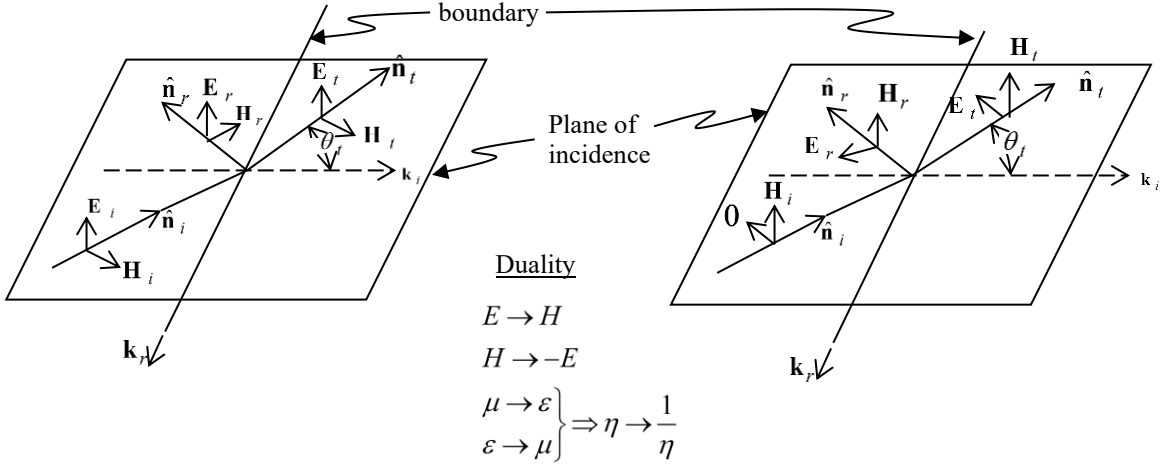
$$\therefore \frac{E_0^2}{2\eta_1} \cos \theta_c - \frac{E_0^2}{2\eta_1} \cos \theta_c = 0 \Rightarrow \text{power conserved between incident \& reflected fields}$$

- Note: (1) no power transferred across the boundary in z - direction
(2) power in region 2 travels $\parallel$ to the boundary.

Appendix A

⊥ polarization

∥ polarization



$$\rightarrow \mathbf{H}_i = H_{i\parallel} e^{-jk_1 \hat{\mathbf{n}}_i \cdot \mathbf{r} \hat{\mathbf{x}}} = \frac{E_{i\parallel}}{\eta_1} e^{-jk_1 \hat{\mathbf{n}}_i \cdot \mathbf{r} \hat{\mathbf{x}}}$$

$$\mathbf{E}_r = E_{r\perp} e^{-jk_1 \hat{\mathbf{n}}_r \cdot \mathbf{r} \hat{\mathbf{x}}} = \Gamma_{\perp} E_{i\perp} e^{-jk_1 \hat{\mathbf{n}}_r \cdot \mathbf{r} \hat{\mathbf{x}}}$$

$$\rightarrow \mathbf{H}_r = H_{r\parallel} e^{-jk_1 \hat{\mathbf{n}}_r \cdot \mathbf{r} \hat{\mathbf{x}}} = \frac{E_{r\parallel}}{\eta_1} e^{-jk_1 \hat{\mathbf{n}}_r \cdot \mathbf{r} \hat{\mathbf{x}}}$$

$$= \frac{\Gamma_{\parallel} E_{i\parallel}}{\eta_1} e^{-jk_1 \hat{\mathbf{n}}_r \cdot \mathbf{r} \hat{\mathbf{x}}}$$

$$\mathbf{E}_t = E_{t\perp} e^{-jk_2 \hat{\mathbf{n}}_t \cdot \mathbf{r} \hat{\mathbf{x}}} = \Gamma_{\perp} E_{i\perp} e^{-jk_2 \hat{\mathbf{n}}_t \cdot \mathbf{r} \hat{\mathbf{x}}}$$

$$\rightarrow \mathbf{H}_t = H_{t\parallel} e^{-jk_2 \hat{\mathbf{n}}_t \cdot \mathbf{r} \hat{\mathbf{x}}} = \frac{E_{t\parallel}}{\eta_2} e^{-jk_2 \hat{\mathbf{n}}_t \cdot \mathbf{r} \hat{\mathbf{x}}}$$

$$= \frac{\Gamma_{\parallel} E_{i\parallel}}{\eta_2} e^{-jk_2 \hat{\mathbf{n}}_t \cdot \mathbf{r} \hat{\mathbf{x}}}$$

$$\mathbf{H}_i = \frac{\hat{\mathbf{n}}_i}{\eta_1} \times (E_{i\perp} e^{-jk_1 \hat{\mathbf{n}}_i \cdot \mathbf{r} \hat{\mathbf{x}}})$$

$$\rightarrow -\mathbf{E}_i = \eta_1 \hat{\mathbf{n}}_i \times H_{i\parallel} e^{-jk_1 \hat{\mathbf{n}}_i \cdot \mathbf{r} \hat{\mathbf{x}}} = \hat{\mathbf{n}}_i \times E_{i\parallel} e^{-jk_1 \hat{\mathbf{n}}_i \cdot \mathbf{r} \hat{\mathbf{x}}}$$

$$\mathbf{H}_r = \frac{\hat{\mathbf{n}}_r}{\eta_1} \times E_{r\perp} e^{-jk_1 \hat{\mathbf{n}}_r \cdot \mathbf{r} \hat{\mathbf{x}}} = \frac{\hat{\mathbf{n}}_r}{\eta_1} \times \Gamma_{\perp} E_{i\perp} e^{-jk_1 \hat{\mathbf{n}}_r \cdot \mathbf{r} \hat{\mathbf{x}}}$$

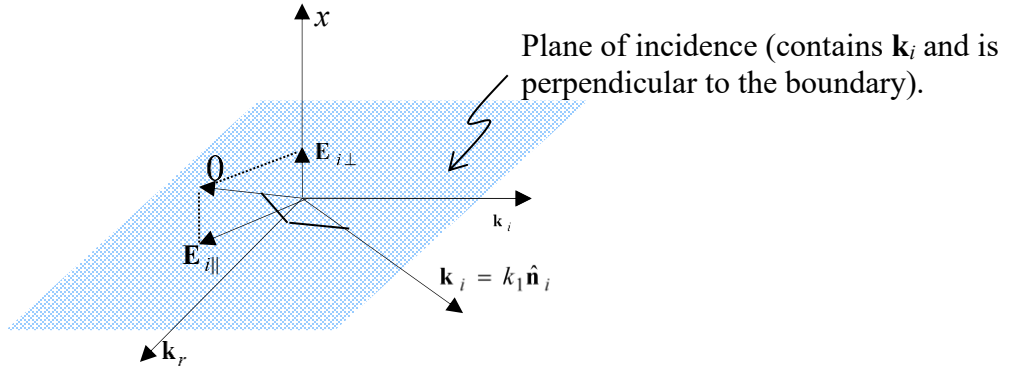
$$\rightarrow -\mathbf{E}_r = \hat{\mathbf{n}}_r \times \Gamma_{\parallel} E_{i\parallel} e^{-jk_1 \hat{\mathbf{n}}_r \cdot \mathbf{r} \hat{\mathbf{x}}}$$

$$\mathbf{H}_t = \frac{\hat{\mathbf{n}}_t}{\eta_2} \times (\Gamma_{\perp} E_{i\perp} e^{-jk_2 \hat{\mathbf{n}}_t \cdot \mathbf{r} \hat{\mathbf{x}}})$$

$$\rightarrow -\mathbf{E}_t = \hat{\mathbf{n}}_t \times \Gamma_{\parallel} E_{i\parallel} e^{-jk_2 \hat{\mathbf{n}}_t \cdot \mathbf{r} \hat{\mathbf{x}}}$$

Where given $\mathbf{E}_i = E_{ix} \hat{x} + E_{iy} \hat{y} + E_{iz} \hat{z}$ then $E_{i\perp} = E_{ix}$ and $E_{i\parallel} = \sqrt{E_{iy}^2 + E_{iz}^2}$.

Now applying these general results to the original problem:



$$\hat{\mathbf{n}}_i = -\sin \theta_i \hat{\mathbf{y}} + \cos \theta_i \hat{\mathbf{z}}$$

$$\hat{\mathbf{n}}_i = -\sin \theta_i \hat{\mathbf{y}} + \cos \theta_i \hat{\mathbf{z}} \text{ so } E_{i\perp} = E_{ix} \text{ and } E_{i\parallel} = \sqrt{E_{iy}^2 + E_{iz}^2}.$$

$\Gamma_{\perp}$, $T_{\perp}$, $\Gamma_{\parallel}$ and $T_{\parallel}$ in this coordinate system become:

$$\Gamma_{\perp} = \frac{\mu_2 k_{1z} - \mu_1 k_{2z}}{\mu_2 k_{1z} + \mu_1 k_{2z}} \quad \Gamma_{\parallel} = \frac{\varepsilon_2 k_{1z} - \varepsilon_1 k_{2z}}{\varepsilon_2 k_{1z} + \varepsilon_1 k_{2z}}$$

$$T_{\perp} = \frac{2\mu_2 k_{1z}}{\mu_2 k_{1z} + \mu_1 k_{2z}} \quad T_{\parallel} = \frac{2\varepsilon_2 \left(\frac{\eta_2}{\eta_1} \right) k_{1z}}{\varepsilon_2 k_{1z} + \varepsilon_1 k_{2z}}$$

Total incident electric field:

$$\begin{aligned} \mathbf{E}_i &= E_{i\perp} e^{-jk_1 \hat{\mathbf{n}}_i \cdot \mathbf{r}} \hat{\mathbf{x}} + \hat{\mathbf{n}}_i \times (E_{i\parallel} e^{-jk_1 \hat{\mathbf{n}}_i \cdot \mathbf{r}} \hat{\mathbf{x}}) \\ &= \underbrace{E_{i\perp}}_{E_{ix}} e^{-jk_1 \hat{\mathbf{n}}_i \cdot \mathbf{r}} \hat{\mathbf{x}} + \underbrace{E_{i\parallel} \cos \theta_i}_{E_{iy}} e^{-jk_1 \hat{\mathbf{n}}_i \cdot \mathbf{r}} \hat{\mathbf{y}} + \underbrace{E_{i\parallel} \sin \theta_i}_{E_{iz}} e^{-jk_1 \hat{\mathbf{n}}_i \cdot \mathbf{r}} \hat{\mathbf{z}} \end{aligned}$$

where

$$\mathbf{E}_i^a = E_{ix} \hat{\mathbf{x}} + E_{iy} \hat{\mathbf{y}} + E_{iz} \hat{\mathbf{z}}$$

Appendix B Interesting form of R&T coefficients

4 FOR $\epsilon_1 > \epsilon_2$ BOTH $\Gamma_{\perp}$ AND $\Gamma_{\parallel}$ ARE COMPLEX, AND $\Gamma_{\perp}$ HAS THE FORM

$$\Gamma_{\perp} = \frac{R + jX}{R - jX} \quad (R, X \text{ REAL})$$

CLEARLY, $|\Gamma_{\perp}| = 1$ (BEING A RATIO OF A COMPLEX NUMBER AND ITS CONJUGATE)

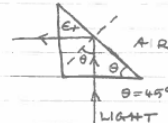
SIMILAR CONCLUSIONS APPLY TO $\parallel$ POLARIZATION (AS LONG AS $k_1 > k_2$).

WELL-KNOWN APPLICATION:

OPTICAL PRISM

$$45^\circ \rightarrow \theta_c = \arcsin \frac{1}{\sqrt{\epsilon_r}}$$

$$\epsilon_r \geq 2$$



TOTAL TRANSMISSION ($\Gamma = 0$?)

$\Rightarrow$ ASSUME NON-MAGNETIC MEDIA ($\mu_1 = \mu_2 = \mu_0$)

THEN,

$$\frac{\sin \theta_c}{\sin \theta_i} = \sqrt{\frac{\epsilon_1}{\epsilon_2}} = \frac{n_2}{n_1}$$

$$\Gamma_{\perp} = \frac{\frac{n_2}{n_1} \cos \theta_i - \cos \theta_c}{\frac{n_2}{n_1} \cos \theta_i + \cos \theta_c} = \frac{\sin(\theta_c - \theta_i)}{\sin(\theta_c + \theta_i)}$$

$$\Gamma_{\parallel} = \frac{\frac{n_2}{n_1} \cos \theta_c - \cos \theta_i}{\frac{n_2}{n_1} \cos \theta_c + \cos \theta_i} = \frac{\tan(\theta_c - \theta_i)}{\tan(\theta_c + \theta_i)}$$

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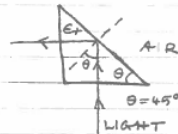
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